

OLIST E-COMMERCE

Customer Analytics & Repeat Purchase Prediction

Exploratory Analysis | RFM Segmentation | Temporal Machine Learning

Final Project Report

Prepared for academic / training submission

1. Executive Summary

This project investigates purchasing behavior, delivery performance, customer satisfaction, historical customer value and repeat purchasing in the Olist Brazilian e-commerce dataset. It combines exploratory analysis, customer segmentation and time-based machine learning to estimate whether an existing customer makes another recorded purchase within 120 days.

The March 1, 2018 development snapshot contains 56,934 existing customers, of whom 503 make another recorded purchase within the prediction horizon (approximately 0.88%). The low observed event rate makes this a difficult ranking problem and motivates the use of PR-AUC and Precision@Top K% rather than accuracy alone.

Final feature experiment	PR-AUC	Precision@Top 1%
Historical RFM	0.02185	5.97%
90-Day RFM	0.02092	5.61%
Historical RFM + Recent Activity	0.02086	6.67%

Historical RFM achieved the highest PR-AUC in this final comparison; adding recent activity yielded the highest precision within the top-ranked 1%. These are retrospective development-set results, not proof of campaign effectiveness or production readiness.

2. Business Understanding

The analysis addresses two connected questions: (1) how operational and delivery characteristics relate to observed customer satisfaction, and (2) which historical customer characteristics help rank customers by subsequent repeat-purchase propensity.

Objectives and evaluation

- Explore regional delivery performance, seller handover, freight, distance and reviews.
- Describe customer purchasing behavior using RFM, segments, cohorts and historical value.
- Predict another purchase within 120 days using information available before each snapshot.
- Compare models, feature sets, temporal periods and activity windows.

The target equals 1 when a customer has a recorded purchase in the 120 days after a snapshot, and 0 otherwise. Primary evaluation uses PR-AUC (average precision), supplemented by ROC-AUC, Precision@Top K% and lift. Ranking customers by purchase propensity is distinct from estimating marketing uplift.

3. Data Understanding and Preparation

Source table	Analytical role
Customers	Unique customer identifiers and location
Orders	Purchases, status and order/delivery timestamps
Order items	Seller links, item prices and freight
Payments	Recorded payments and spending measures
Reviews	Customer review scores and answer timestamps
Sellers / geolocation	Seller locations and approximate geographical distance
Products	Purchased product categories

Order-level joins require care: one order may have several items or payments. Customer-level modeling uses `customer_unique_id`, whereas `customer_id` identifies the order-associated customer record. Timestamps are converted to `datetime`; order-level measures are aggregated before customer-level summaries where appropriate.

For every prediction snapshot, purchase features use only earlier purchases. Delivery experience is included only if delivery was completed before the cutoff; reviews are included only if answered before the cutoff. The label uses purchases in the subsequent 120-day window. These choices are intended to reduce temporal leakage.

4. Exploratory Data Analysis

Delivery performance and customer reviews

Late delivery is defined by actual delivery occurring after the estimated delivery date. The analysis examines delivery duration and lateness across customer states, seller groups and shipping-distance bands, and compares review scores for on-time and late orders. These comparisons establish descriptive associations; they do not independently establish causal effects on ratings.

Seller handover and logistics

Seller processing time is approximated from approval to carrier handover; downstream delivery time is measured from handover to customer delivery. A four-way classification distinguishes on-time versus late handover and on-time versus late final delivery. An order-level handover timestamp cannot always identify each seller's individual behavior for multi-seller orders.

Geographical distance

Representative ZIP-prefix coordinates are used to estimate straight-line seller-to-customer distance with the Haversine formula. Distance bands support exploratory comparisons of freight, delivery duration and lateness. Straight-line distances do not represent road networks, transport routes or actual logistics costs.

Operational interpretation

Useful monitoring outputs include late-delivery rate by state, seller handover rates, freight by distance, and review distributions by delivery outcome. Exact regional and seller-level numerical conclusions should be inserted only after checking the corresponding saved analysis outputs; they are not asserted in this report.

5. Customer Analytics

RFM and segmentation

Customers are described by recency (days since last recorded purchase), frequency (number of recorded orders) and monetary value (historical spending). RFM scoring supports descriptive segments such as Champions, Loyal Customers, At Risk, Repeat Customers, High Value New, Recent Customers, Inactive Customers and Regular Customers. Segment sizes and spending shares should be interpreted relative to the available observation window.

Cohort analysis

Customers are grouped by their first recorded purchase month. Subsequent monthly purchasing activity is examined relative to each cohort's initial customer count. Recent cohorts have shorter follow-up, so unobserved later months must not be treated as zero retention.

Historical value versus CLV

Historical customer value sums recorded spending; average order value divides historical spending by recorded order count. Neither is a complete lifetime value measure. The predictive model estimates the occurrence of a future purchase, not future revenue, margin or predictive customer lifetime value.

6. Predictive Modeling and Experiments

6.1 Temporal design and features

The initial model comparison trained on July 1 and October 1, 2017 snapshots and evaluated on March 1, 2018. Historical features include recency, frequency, monetary value, average order value, tenure and average interpurchase interval. Extended groups add one-time status, purchases in the preceding 30/90 days, reviews, late delivery, delivery duration, freight, product diversity and availability indicators. A customer can contribute multiple customer-snapshot observations.

6.2 Algorithms and metrics

Logistic Regression (with scaling and class weighting), Random Forest and XGBoost were compared. The observed positive rate in the March 2018 development population is about 0.88%; a trivial all-negative classifier would therefore have high accuracy while identifying no returners. PR-AUC and top-of-ranking precision are more informative for the stated ranking use case.

6.3 Feature ablation: Logistic Regression

Feature group	PR-AUC	ROC-AUC	Returning customers in Top 500
RFM only	0.0326	0.5958	34
Purchase behavior	0.0193	0.5990	32
Purchase + experience	0.0193	0.6046	33

In this ablation, the RFM-only configuration had the highest PR-AUC, whereas the extended configuration had the highest ROC-AUC. These metrics emphasize different aspects of ranking. The ablation values should not be treated as interchangeable with differently engineered RFM variants in later experiments.

6.4 Temporal robustness

Test snapshot	Test customers	Positive rate	RFM PR-AUC	Extended PR-AUC
2018-01-01	43,529	0.93%	0.0256	0.0181
2018-03-01	56,934	0.88%	0.0325	0.0192
2018-05-01	70,590	0.79%	0.0228	0.0218

RFM produced higher PR-AUC in all three reported periods, but the difference narrowed in May 2018. The changing values demonstrate sensitivity to evaluation period. Training snapshots and historical coverage also changed across these experiments.

6.5 Training-population activity windows

Eligible training customers	Train observations	PR-AUC	Precision@Top 1%
All historical customers	46,197	0.01923	5.97%
Purchased within 180 days	40,406	0.02309	6.14%
Purchased within 90 days	26,079	0.02619	5.09%

The 90-day training-eligibility rule produced the highest PR-AUC, while 180-day eligibility produced the highest Precision@Top 1%. This experiment filters training observations; it is different from shortening the history used to calculate features.

6.6 RFM feature-observation windows

RFM history used	PR-AUC	Precision@Top 1%
Previous 90 days	0.02092	5.61%
Previous 180 days	0.02049	5.79%
All history (window experiment)	0.01923	5.97%

Shorter RFM history improved PR-AUC in this window-specific comparison, but all-history features achieved higher top-1% precision. These rows come from the observation-window experiment and should not be

conflated with the separate final feature-combination experiment.

6.7 Final feature-combination comparison

Configuration	PR-AUC	Precision@1%	Returning in Top 1%	Precision@ 5%
Historical RFM	0.02185	5.97%	34	2.42%
90-Day RFM (context)	0.02092	5.61%	32	2.39%
Historical RFM + recent activity	0.02086	6.67%	38	2.49%

The final comparison evaluated 56,934 customers, including 503 returners. Historical RFM achieved higher PR-AUC, while the combined historical-plus-recent feature set placed four more returners in the highest-ranked 1% than historical RFM alone (38 versus 34). This small difference is not evidence of a stable improvement without additional independent temporal testing.

7. Business Recommendations

Delivery monitoring

Track late delivery by region, seller and delivery stage; investigate unusual patterns before attributing causes.

Seller performance

Compare handover timeliness alongside order volume and operational context, avoiding simplistic rankings from small samples.

RFM-based customer research

Use descriptive segments to formulate differentiated customer hypotheses, rather than assuming any segment is automatically responsive to incentives.

Controlled retention experiments

If using propensity rankings for campaigns, randomize treatment and control groups; evaluate incremental orders, revenue, costs and profitability.

Operational dashboard

Combine customer segments, cohort purchasing, delivery metrics and predictive performance in a regularly refreshed dashboard.

8. Limitations, Future Work and Conclusion

Limitations

- Observed customer histories are limited to the available marketplace data and observation period.
- The 120-day repeat event is rare, and absolute predictive performance is limited.
- Some customers appear in multiple snapshot observations; results vary over time.
- Approximate geographic distances and order-level logistics timestamps limit operational attribution.

- Propensity is not causal uplift, predictive CLV or a guarantee of profitable marketing action.

Future work

Evaluate on further non-overlapping future periods; assess calibration and ranking stability; test alternative sequence and category features; estimate future revenue separately from purchase occurrence; and run randomized campaigns before estimating intervention effectiveness. Uplift modeling would require appropriate treatment and control data.

Conclusion

This project provides a reproducible framework for exploring e-commerce operations and customer behavior and for retrospectively evaluating repeat-purchase ranking models. Simple historical purchasing variables were informative, while additional features and alternative activity windows changed performance in metric-dependent ways. The findings support further validation and experimentation, not claims of proven retention uplift or deployment readiness.

Reporting note

Numerical tables reproduce the results discussed in the project conversation. The underlying code outputs, source CSVs and charts should be attached or cross-checked before formal submission. No unsupported numerical EDA findings or unverified external references have been added.

9. HVIA Task Requirements: Company Research and Solution Proposal

9.1 Company Research: Olist

Olist is a Brazilian e-commerce business associated with a marketplace ecosystem that connects merchants, customers, orders, and delivery partners. The public Olist dataset contains linked order, customer, seller, product, payment, review, and geographical records. These records make it possible to study the customer journey from purchase through fulfillment and post-purchase feedback. This is a dataset-based understanding of the business context, not an assessment of Olist's current operating model.

The commercial questions explored in this project concern delivery reliability, seller fulfillment, customer experience, purchasing frequency, and repeat-purchase propensity. The available data does not include full marketing exposure, acquisition costs, margins, or complete customer lifetimes; recommendations must therefore be tested before business impact can be claimed.

9.2 Proposed HVIA Data / AI Solution

Solution concept: Customer & Delivery Intelligence Platform — a decision-support product combining operational monitoring, customer analytics, and a repeat-purchase scoring service.

Component	Practical implementation
Operational dashboard	Monitor delivery lateness, seller handover, geography, freight and review patterns; allow drill-down by time and segment.
Customer intelligence	Show RFM segments, cohort purchasing activity, historical value and customer-level behavioral indicators.
Propensity scoring	Generate time-stamped 120-day repeat-purchase scores and monitor PR-AUC and Precision@Top K% across future snapshots.
Experimentation layer	Randomly test selected customer-contact strategies against holdout groups; report incremental purchases, revenue and campaign costs.

Delivery plan: Begin with data-quality checks and a descriptive dashboard; validate the scoring model on additional future periods; run a limited, randomized pilot before any scaled targeting. Success criteria should include data reliability, stable ranking performance, operational usability and measured incremental business value.

Important limitation: The present model predicts observed repeat purchases, not the causal effect of outreach. The current data cannot substantiate uplift or return on investment without an intervention study.

9.3 Outreach Draft — LinkedIn / Email

Subject: Data & AI opportunity: Olist customer and delivery intelligence

Hello [Name],

I recently completed an analysis of the public Olist e-commerce dataset, covering delivery performance, customer segmentation, and 120-day repeat-purchase prediction. The work highlights opportunities to bring operational and customer indicators together in a practical decision-support dashboard.

I would welcome the opportunity to share a short concept for an HVIA-supported Customer & Delivery Intelligence Platform, including seller and delivery monitoring, customer analytics, and carefully validated propensity scoring. Any marketing application would be tested through controlled experiments before making impact claims.

Would you be open to a brief conversation to discuss whether this type of solution aligns with your team's priorities?

Best regards,

[Your Name]

[Your Contact / LinkedIn]

10. Submission and Evidence Notes

This report summarizes the project and its reported experimental outputs. Supporting work should include the executable analysis code or notebook, the relevant charts, and (where permitted) instructions for obtaining the source dataset. Figures and exact operational findings not included in this PDF should not be treated as independently verified results.

Before submitting: insert your name and contact details in the outreach draft; confirm numerical tables against your own code outputs; attach the code/notebook and actual charts; and ensure any Drive or GitHub link has appropriate viewing permissions.

Report scope: historical public Olist dataset; results are retrospective and should not be presented as Olist's current business performance.